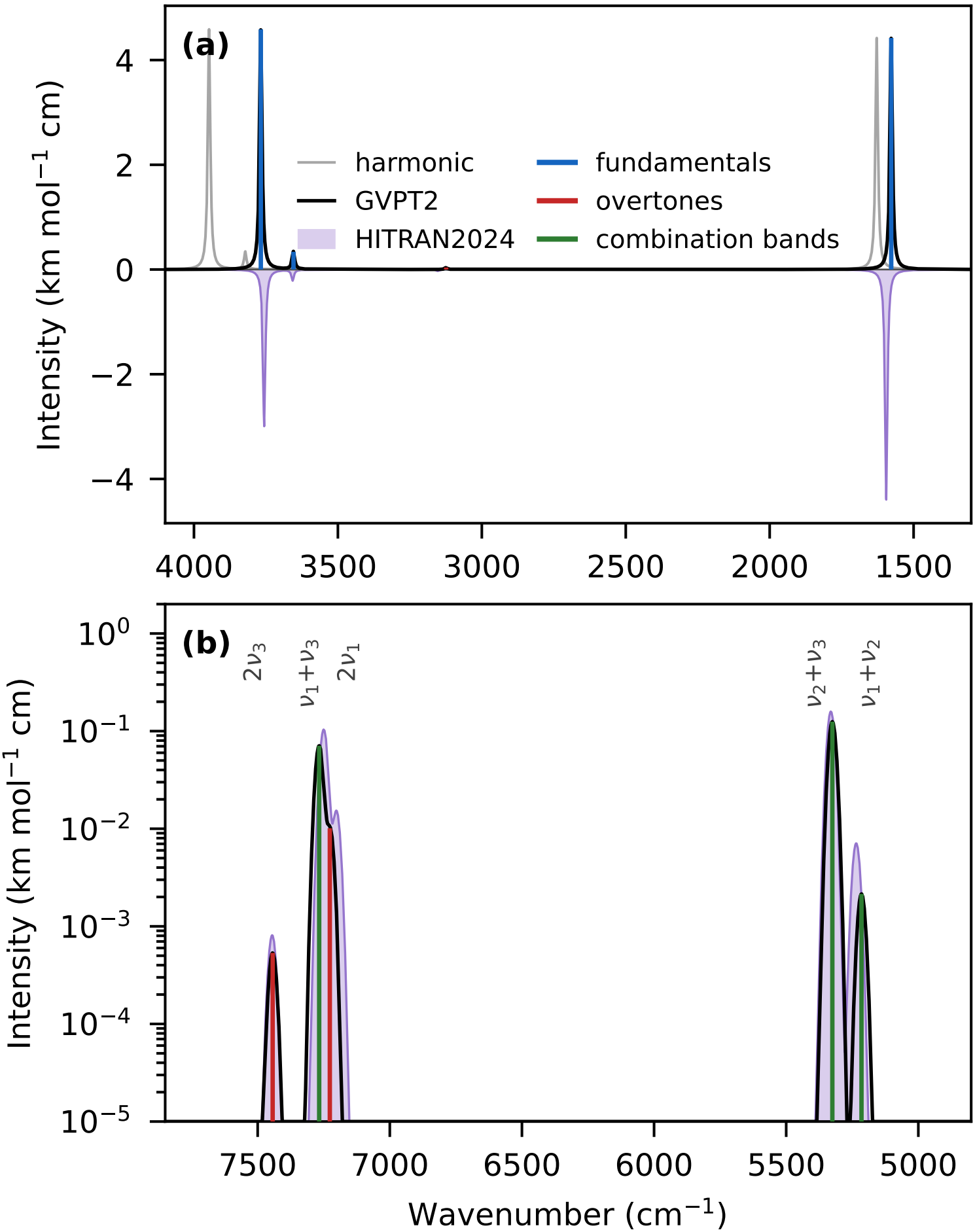

(a)
harmonic
GVPT2
HITRAN2024
fundamentals
overtones
combination bands
Intensity (km mol^{-1} cm)
4
2
0
−2
−4
4000
3500
3000
2500
2000
1500
(b)
$2\nu_3$
$\nu_1+\nu_3$
$2\nu_1$
$\nu_2+\nu_3$
$\nu_1+\nu_2$
Intensity (km mol^{-1} cm)
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
7500
7000
6500
6000
5500
5000
Wavenumber (cm^{-1})